

Adaptive Fuzzy Neural Network Command Filtered Impedance Control of Constrained Robotic Manipulators With Disturbance Observer

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Abstract—This article proposes an adaptive fuzzy neural network (NN) command filtered impedance control for constrained robotic manipulators with disturbance observers. First, barrier Lyapunov functions are introduced to handle the full-state constraints. Second, the adaptive fuzzy NN is introduced to handle the unknown system dynamics and a disturbance observer is designed to eliminate the effect of unknown bound disturbance. Then, a modified auxiliary system is designed to suppress the input saturation effect. In addition, the command filtered technique and error compensation mechanism are used to directly obtain the derivative of the virtual control law and improve the control accuracy. The barrier Lyapunov theory is used to prove that all the signals in the closed-loop system are semiglobally uniformly ultimately bounded. Finally, simulation studies are performed to illustrate the effectiveness of the proposed control method.

Index Terms—Command filter, disturbance observer, full-state constraints, fuzzy neural network (NN), impedance control, input saturation.

I. INTRODUCTION

IN RECENT years, impedance control has been proposed and widely used as an effective method to regulate the interaction force between the end of the robotic manipulator and the environment. Hogan [1] first proposed the design method of impedance controller to adjust the interaction forces. In [2], two adaptive impedance control algorithms are proposed to describe the ideal dynamic relationship between the robotic manipulator and the workspace. However, all the above-mentioned methods are based on robot dynamics are known. For actual applications, there are many uncertain dynamics and unknown disturbances [3]–[6], and there is hysteretic coupling between the actuator and the links in a

manipulator [7], [8], which brings certain difficulties to the controller design. The greater challenge is the state constraints and input saturation [9] caused by the limitations of the robotic manipulator working environment and hardware performance. Ignoring these constraints may result in reduced robot performance or even hardware damage, and more seriously, may lead to casualties. Therefore, it is necessary to design an impedance controller which can effectively deal with the uncertain dynamics and unknown time-varying disturbances in the practical application of robotic manipulators, realize the full-state constraints control and compensate for the input saturation effect.

In order to solve the influence of uncertain dynamics, adaptive NN control is proposed as a kind of control method with learning ability [10]–[12], which can be used to effectively approximate nonlinear functions in nonlinear system control. In [13], Huang *et al.* proposed an adaptive impedance control method based on model-free NN for constrained robots. In [14], an adaptive NN control method is proposed for uncertain stochastic nonlinear systems appearing dead-zone. In some researches, fuzzy technology is introduced into the structure of adaptive NN to form fuzzy NN, which can not only avoid the lack of interpretability of neural networks (NNs) but also improve the learning ability of fuzzy systems. In [15], an adaptive fuzzy impedance control strategy is proposed based on NN visual servo in an unknown environment. In [16], fuzzy NN control is studied to suppress the flexible vibration maximally and realize accurate tracking of the desired trajectory.

For deducing the unknown disturbance from the external environment without additional sensor information, disturbance observer has been widely used [17], [18]. In [17], a nonlinear disturbance observer is designed to estimate the unknown disturbance in the nonlinear system. In [18], the disturbance observer is added to the fuzzy logic full-state constraints control to compensate for the unknown disturbance.

For achieving state constraints, barrier Lyapunov functions (BLFs) have been considered to be an effective tool [19]. He *et al.* [20] proposed an adaptive NN full-state constraints control method based on the logarithmic BLFs. Ding *et al.* [21] proposed NNs-based adaptive full-state constraint control method for wheeled mobile robots. Another challenge of constraints control design is input saturation, which seriously limits the system performance and even leads to system instability. For designing control methods that can effectively deal with input saturation, several research results have been

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proposed. In [22] and [23], the influence of input saturation is eliminated by designing the auxiliary system, but it does not deal with the control input nonsmooth problem caused by input saturation, and the state of the auxiliary system cannot converge to zero in any state. Wen *et al.* [24] used a smooth function to approximate the saturation and Nussbaum gain technique to compensate the nonlinearity caused by input saturation. In [25], introducing an auxiliary design system to handle the input saturation and saturation effect can be properly handled in the stability analysis. In [24] and [25], the hypertangent function is used to replace the saturated control law function to obtain a smooth control law, and the difference between the control law and the hypertangent function is used to drive the auxiliary system. However, as long as the control law is not equal to zero, there is always a difference between the hypertangent function and the control law; therefore, even if there is no input saturation, the state of the auxiliary system will not converge to zero, which may degrade the control performance.

Furthermore, the above-mentioned control designs are based on the backstepping technique, which may increase the online calculation burden due to the derivative of the virtual control law [26], [27]. For solving this problem, this article adds command filtered control (CFC) [28], [29] technique with compensation mechanism to the impedance control of robotic manipulators to directly obtain the derivative of the virtual control law, reduce the burden of online calculation, and improve the accuracy of the control system. Up to now, it is still a challenging task to combine CFC technology with full-state constrained impedance control of constrained manipulators.

Based on the observations earlier, this article proposes an adaptive fuzzy NN command filtered impedance control of constrained robotic manipulators with disturbance observers. The primary contributions can be summarized as follows.

- 1) The CFC technique with compensation mechanism is designed to directly obtain the derivative of the virtual control law, which can reduce the burden of online calculation and improve the control accuracy. It is the first time that the command filtering full-state constrained impedance control is used for constrained robotic manipulators with input saturation.
- 2) Combined with CFC and full-state constraint control technique, a modified auxiliary system is introduced to deal with the effect of input saturation. Compared with [22] and [23], the modified auxiliary system can obtain a smooth control input law, and the state of the auxiliary system can be zero. Compared with [24] and [25], the state of the modified auxiliary system proposed in this article will converge to zero if the control input saturation does not occur, which overcomes the defect in [24] and [25].
- 3) The adaptive fuzzy NN is adopted to approximate the unmodeled dynamics. And combined with CFC technology and auxiliary system, a disturbance observer is designed to compensate for the unknown time-varying disturbance.

II. MATHEMATICAL MODEL AND PRELIMINARIES

A. System Description

Consider an m -link robotic manipulator system with its unknown environment, and force sensors mounted on the end-effector are used to detect the force exerted by the environment. The dynamics model [30] in the Cartesian space is described as

$$D_x(q)\ddot{x} + C_x(q, \dot{q})\dot{x} + G_x(q) = S(\tau) - \tau_d - \tau_e \quad (1)$$

where $q, \dot{q}, \ddot{q} \in \mathbb{R}^m$ are the joint angular, displacement, and velocity, $x, \dot{x}, \ddot{x} \in \mathbb{R}^n$ are the position, velocity, and acceleration vectors of the robotic manipulator's end-effector in the Cartesian space, $D_x(q) \in \mathbb{R}^{n \times n}$ denotes the symmetric positive definite Cartesian inertia matrix, which is symmetric and positive definite, $C_x(q, \dot{q}) \in \mathbb{R}^{n \times n}$ denotes the Cartesian Coriolis and centrifugal matrix, $G_x(q) \in \mathbb{R}^n$ denotes the gravitational force vector of the manipulator in Cartesian space, $\tau \in \mathbb{R}^n$ denotes the control input vector, $S(\tau) \in \mathbb{R}^n$ denotes the saturation function to the control input vector τ , $\tau_d \in \mathbb{R}^n$ denotes the unknown disturbance moment vector of the external environment, and $\tau_e \in \mathbb{R}^n$ denotes the vector of constraining force on robotic manipulator's end-effector in Cartesian space. n and m denote the dimension of the workspace and the degrees of freedom (DOF), respectively.

The relationship between $\dot{x}$, $\ddot{x}$ and $\dot{q}$, $\ddot{q}$ can be expressed as

$$\dot{x} = J\dot{q}, \quad \ddot{x} = \dot{J}\dot{q} + J\ddot{q} \quad (2)$$

where J is the Jacobian matrix assumed to be known and nonsingular. When $m > n$, J^{-1} should be replaced by the pseudo-inverse of J , which is represented as J^+ and defined as follows:

$$J^+ = J^T(JJ^T)^{-1}. \quad (3)$$

Let $x_1 = x = [x_{11}, \dots, x_{1n}]^T$ and $x_2 = \dot{x}_1 = [x_{21}, \dots, x_{2n}]^T$. According to (1), the robotic manipulator dynamics could be expressed as follows:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = D_x^{-1}(q)[S(\tau) - \tau_d - \tau_e - C_x(q, \dot{q})x_2 - G_x(q)]. \end{cases} \quad (4)$$

In practical applications, $D_x(q)$, $C_x(q, \dot{q})$, and $G_x(q)$ in (4) are difficult to obtain accurately. The desired impedance dynamics [31], [32] between τ_e and e in the workspace is describe as

$$M_d\ddot{e} + B_d\dot{e} + K_de = \tau_e \quad (5)$$

where $e = x_d - x$, and $x_d = [x_{d1}, \dots, x_{dn}]^T \in \mathbb{R}^n$ denotes the commanded trajectories, which is bounded and twice differentiable. M_d , B_d , and $K_d \in \mathbb{R}^{n \times n}$ are specified by the user. M_d , B_d , and K_d denote the desired inertia, damping, and stiffness matrices, respectively. There has $x_d = x$ and $\tau_e = 0$, if the robotic manipulator is moving in free space. If x tracks the desired trajectory $x_r \in \mathbb{R}^n$ precisely, (5) can be written as

$$M_d(\ddot{x}_d - \ddot{x}_r) + B_d(\dot{x}_d - \dot{x}_r) + K_d(x_d - x_r) = \tau_e. \quad (6)$$

Therefore, x_r can be obtained from (6) if x_d , M_d , C_d , G_d , and the force τ_e are given.

Assumption 1: The robotic manipulator operation region Ω_x does not contain kinematic singularities, which means J is nonsingular as long as $x(t) \in \Omega_x, \forall t \geq 0$.

Assumption 2: The commanded trajectory $x_d(t) \in \Omega_x, \forall t \geq 0$, and its first derivative $\dot{x}_d(t)$ and second derivative $\ddot{x}_d(t)$ are smooth, bounded, and known with $\|x_d(t)\| \leq Y_0 < k_{c1}$, $\|\dot{x}_d(t)\| \leq Y_1$, and $\|\ddot{x}_d(t)\| \leq Y_2$, where Y_0, Y_1 , and Y_2 are positive constants.

Assumption 3: The unknown disturbance τ_d and its first derivative $\dot{\tau}_d$ are smooth and bounded.

B. Input Saturation

$S(\tau) = [S_1(\tau_1), \dots, S_n(\tau_n)]^T$ can be expressed as [24]

$$S_i(\tau_i) = \text{sat}(\tau_i) = \begin{cases} \tau_{i \max}, & \tau_i > \tau_{i \max} \\ \tau_i, & \tau_{i \min} \leq \tau_i \leq \tau_{i \max} \\ \tau_{i \min}, & \tau_i < \tau_{i \min} \end{cases} \quad (7)$$

where $\tau_{i \max} > 0$ and $\tau_{i \min} < 0$ are the maximum and minimum input of the actuator. When $\tau_i = \tau_{i \max}$ and $\tau_i = \tau_{i \min}$, two sharp corners will be generated. For resolving this problem, defined $h_i(\tau_i)$ as follows:

$$h_i(\tau_i) = \begin{cases} \mu_i \tau_{i \max} + \bar{\tau}_{i \max}, & \tau_i > \mu_i \tau_{i \max} \\ \tau_i, & \mu_i \tau_{i \min} \leq \tau_i \leq \mu_i \tau_{i \max} \\ \mu_i \tau_{i \min} + \underline{\tau}_{i \min}, & \tau_i < \mu_i \tau_{i \min} \end{cases} \quad (8)$$

where $\mu_i \in (0, 1)$ is defined by user, and $\bar{\tau}_{i \max}$ and $\underline{\tau}_{i \min}$ are defined as

$$\begin{aligned} \bar{\tau}_{i \max} &= (1 - \mu_i) \tau_{i \max} \tanh\left(\frac{\tau_i - \mu_i \tau_{i \max}}{(1 - \mu_i) \tau_{i \max}}\right) \\ \underline{\tau}_{i \min} &= (1 - \mu_i) \tau_{i \min} \tanh\left(\frac{\tau_i - \mu_i \tau_{i \min}}{(1 - \mu_i) \tau_{i \min}}\right). \end{aligned} \quad (9)$$

Remark 1: It is notice that piecewise function $h_i(\tau_i)$ is twice differentiable relative to τ . Then, $S_i(\tau_i)$ can be rewritten as $S_i(\tau_i) = h_i(\tau_i) + Y_i(\tau_i)$, where $Y_i(\tau_i)$ is a bounded function. Define $\Delta\tau = S(\tau) - \tau$, since the system is controllable, $\Delta\tau$ should be bounded.

C. Fuzzy Neural Networks

Consider l fuzzy IF-THEN rules $\mathbb{R}^{(k)}$: If x_1 is $A_1^k, \dots$ and x_n is A_n^k . Then, y is W^k , $k = 1, \dots, l$, where $\mathbb{R}^{(k)}$ denotes the k th rule, $1 \leq k \leq l$. $(x_1, x_2, \dots, x_n)^T \in \mathbb{U} \subset \mathbb{R}^n$ and $y \in \mathbb{R}$ are the linguistic variables that are associated with the inputs and output of the fuzzy NNs, respectively. A_i^k denotes the fuzzy sets characterized by fuzzy membership functions $\mu_{A_i^k}(x_i)$ in $\mathbb{U}$, and W^k denotes the fuzzy sets characterized by fuzzy membership functions $\mu_{W^k}(y)$ in $\mathbb{R}$. The results are as follows [33]:

$$y(x) = \frac{\sum_{k=1}^l \left(\prod_{i=1}^n \mu_{A_i^k}(x_i) \right) y_k}{\sum_{k=1}^l \left(\prod_{i=1}^n \mu_{A_i^k}(x_i) \right)} \quad (10)$$

where $x = [x_1, \dots, x_n]^T$, y^k is the point at $\mu_{W^k}(y^k) = 1$ and $\mu_{A_i^k}$ denotes as a Gaussian form $\mu_{A_i^k} = \exp[-(x_i - c_{ik})^2 / \sigma_{ik}^2]$, where c_{ik} denotes the central value, and σ_{ik} denotes the width of the Gaussian.

In addition, the fuzzy basis function vector and weight vector are defined as $\phi(x) = [\phi_1, \phi_2, \dots, \phi_l]^T$ and $\theta = [y_1, y_2, \dots, y_l]^T$, respectively, where

$$\phi_k = \left(\prod_{i=1}^n \mu_{A_i^k}(x_i) / \sum_{k=1}^l \left(\prod_{i=1}^n \mu_{A_i^k}(x_i) \right) \right), 1 \leq k \leq l.$$

Hence, (10) can be described as

$$y = \theta^T \phi(x). \quad (11)$$

Through the above-mentioned analysis, it can be concluded that fuzzy NNs can approximate any known continuous function $f_i(x_i)$, $i = 1, 2, \dots, n$ to arbitrary accuracy on a compact set Ω . Therefore, there is a $\theta_i^{*T} \phi(x_i)$ in the fuzzy NN for any constant $\delta_i > 0$ that makes

$$f_i(x_i) = \theta_i^{*T} \phi(x_i) + \varepsilon_i$$

where θ_i^* is an unknown constant parameter vector, $\phi(x_i)$ is the fuzzy basis function, and ε_i is the approximation error, which satisfies $\max_{x \in \Omega} \|\varepsilon_i\| \leq \delta_i$, where $\delta_i > 0$ is unknown bound.

D. Preliminaries

Lemma 1 [34]: For bounded initial conditions, if there exists a C^1 continuous and positive definite Lyapunov function $V(x)$ satisfying $\kappa_1(\|x\|) \leq V(x) \leq \kappa_2(\|x\|)$ such that $\dot{V}(x) \leq -\rho V(x) + c$, where $\kappa_1, \kappa_2: \mathbb{R}^n \rightarrow \mathbb{R}$ are class κ functions, and ρ and c are two positive constants, then the solution $x(t)$ is uniformly bounded.

Lemma 2 [35]: For any positive constant k , the following inequality holds for any $x \in \mathbb{R}$ in the interval $|x| < k$:

$$\log \frac{k^2}{k^2 - x^2} \leq \frac{x^2}{k^2 - x^2}.$$

Lemma 3 [36]: For real variables o and c , the following inequality holds for any positive constants p, q , and l :

$$|o|^p |c|^q \leq \frac{p}{p+q} l |o|^{p+q} + \frac{q}{p+q} l^{\frac{-p}{q}} |c|^{p+q}.$$

Lemma 4 [29]: The definition of the command filters are displayed as follows:

$$\begin{cases} \dot{\eta}_1 = \omega_n \eta_2, \\ \dot{\eta}_2 = -2\xi \omega_n \eta_2 - \omega_n (\eta_1 - \alpha_1) \end{cases}$$

where η_1 and η_2 are both command filter output signals. And if the input signal α_1 satisfies $|\dot{\alpha}_1| < \rho_1$ and $|\ddot{\alpha}_1| < \rho_2$ for all $t \geq 0$, where ρ_1 and ρ_2 are positive constants, and $\eta_1(0) = \alpha_1(0)$ and $\eta_2(0) = 0$ then for any $\beta > 0$, there exists $\xi \in (0, 1]$ and $\omega_n > 0$ such that $|\eta_1 - \alpha_1| \leq \beta$, and $|\dot{\eta}_1|$, $|\ddot{\eta}_1|$, and $|\ddot{\eta}_2|$ are bounded.

III. CONTROL DESIGN

Define error variables z_1, z_2, v_1 , and v_2 as follows:

$$\begin{aligned} z_1 &= x_1 - x_r = [z_{11}, \dots, z_{1n}]^T \\ z_2 &= x_2 - x_{1,c} = [z_{21}, \dots, z_{2n}]^T \\ v_1 &= z_1 - \xi_1 = [v_{11}, \dots, v_{1n}]^T \\ v_2 &= z_2 - \xi_2 - \varphi = [v_{21}, \dots, v_{2n}]^T \end{aligned} \quad (12)$$

where $x_{1,c} = [x_{1,c1}, \dots, x_{1,cn}]^T$ is output signal of the command filter when virtual control law $\alpha = [\alpha_1, \dots, \alpha_n]^T$ is the control input signal, $\xi_i = [\xi_{i1}, \dots, \xi_{in}]^T$, $i = 1, 2$, is the error compensation signals, and $\varphi = [\varphi_1, \dots, \varphi_n]^T$ is the auxiliary system. When the end of the robotic manipulator moves in the free space, $\tau_e = 0$, the desired trajectory x_r is consistent with the command trajectory x_d ; when the end of the robotic manipulator is in contact with the external environment, $\tau_e \neq 0$ can be measured by the force sensor, and the required desired trajectory x_r can be obtained by impedance dynamics (6). The virtual control law α , error compensation signals ξ_i , and auxiliary system φ will be given in the following design. Define compact sets as $\Omega_{v1} := \{|v_{1i}| < k_{ai}, i = 1, 2, \dots, n\}$, $\Omega_{v2} := \{|v_{2i}| < k_{bi}, i = 1, 2, \dots, n\}$, $\Omega_{x1} := \{|x_{1i}| < k_{ci}, i = 1, 2, \dots, n\}$ and $\Omega_{x2} := \{|x_{2i}| < k_{ci}, i = 1, 2, \dots, n\}$.

For the convenience, D , C , and G are used to represent $D_x(q)$, $C_x(q, \dot{q})$, and $G_x(q)$, respectively.

Consider that there are uncertainties in D , C , and G , define D_k , C_k , and G_k to represent the known part in the dynamic model, D_u , C_u , and G_u to represent other unknown parts, and the dynamic relationship can be described as

$$D = D_k + D_u, C = C_k + C_u, G = G_k + G_u. \quad (13)$$

According to the dynamics model (4) combined with equation (12) and (13), use D_I represents D_k^{-1} , and $D_I = [D_{I1}, D_{I2}, \dots, D_{In}]^T$, the following equations can be obtained:

$$\begin{aligned} \dot{v}_1 &= z_2 + x_{1,c} - \dot{x}_r - \dot{\xi}_1 \\ \dot{v}_{1i} &= z_{2i} + x_{1,ci} - \dot{x}_{ri} - \dot{\xi}_{1i} \\ \dot{v}_2 &= D_I[h(\tau) + Y(\tau) - G_k - D_k \dot{x}_{1,c} - \tau_d - \tau_e - C_k x_2] \\ &\quad - f(Z) - \dot{\xi}_2 - \dot{\varphi} \\ \dot{v}_{2i} &= D_{Ii}[h(\tau) + Y(\tau) - G_k - D_k \dot{x}_{1,c} - \tau_d - \tau_e - C_k x_2] \\ &\quad - f_{ii}(Z) - \dot{\xi}_{2i} - \dot{\varphi}_i \end{aligned} \quad (14)$$

where $h(\tau) = [h_1(\tau_1), \dots, h_n(\tau_n)]^T$, $Y(\tau) = [Y_1(\tau_1), \dots, Y_n(\tau_n)]^T$, and $f(Z) = [f_1(Z), \dots, f_n(Z)]^T = D_I(D_u \dot{x}_{1,c} + C_u x_2 + G_u + D_u \dot{z}_2)$, and $Z = [x_1^T, x_2^T, \dot{z}_2^T, \dot{x}_{1,c}^T]^T$. According to the adaptive fuzzy NN, for any ε_2 , it can be known that $f(Z) = \theta_2^{*T} \phi_2(Z) + \delta_2(Z)$ with the approximation error $\delta_2(Z)$ satisfying $|\delta_2(Z)| \leq \varepsilon_2$, where $\theta_2^* = [\theta_{21}^*, \dots, \theta_{2n}^*]^T$ and $\phi_2(Z) = [\phi_{21}, \dots, \phi_{2n}]^T$.

Design the following auxiliary system:

$$\dot{\varphi} = D_I[-K_\eta \varphi + h(\tau) - \tau] \quad (15)$$

where $K_\eta = \text{diag}\{k_{\eta 1}, \dots, k_{\eta n}\} > 0$. The following inequality can be obtained:

$$\begin{aligned} \dot{v}_{2i} &\leq D_{Ii}[\tau + Y(\tau) - G_k - D_k \dot{x}_{1,c} - \tau_d - \tau_e - C_k x_2 + K_\mu \varphi] \\ &\quad - \theta_{2i}^{*T} \phi_{2i}(Z) - \varepsilon_{2i} - \dot{\xi}_{2i}. \end{aligned} \quad (16)$$

The overall control block diagram of the proposed scheme is shown in Fig. 1.

Design virtual control law α , the control input vector $\tau = [\tau_1, \tau_2, \dots, \tau_n]^T$, compensation signals ξ_j , ($j = 1, 2$) and the

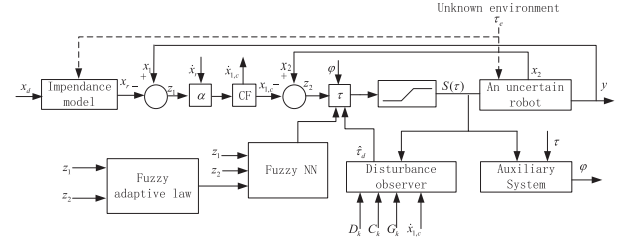


Fig. 1. The overall control block diagram of the proposed method.

adaptive laws $\hat{\theta}_{2i}$ as follows:

$$\alpha = -K_1 z_1 + \dot{x}_r \quad (17)$$

$$\begin{aligned} \tau &= D_k(-K_2 z_2 - A + \hat{\theta}_2^T \phi_2 + \dot{x}_{1,c} - \frac{3}{4} K_{v2}) \\ &\quad + \tau_e + \hat{\tau}_d + C_k x_2 + G_k - K_\eta \varphi \end{aligned} \quad (18)$$

$$\dot{\xi}_1 = -K_1 \xi_1 + \xi_2 + \varphi + (x_{1,c} - \alpha) \quad (19)$$

$$\dot{\xi}_2 = -K_2 (\xi_2 + \varphi) \quad (20)$$

$$\dot{\hat{\theta}}_{2i} = -r_{2i} \left(\frac{\phi_{2i} v_{2i}}{k_{bi}^2 - v_{2i}^2} + \sigma_{2i} \hat{\theta}_{2i} \right), i = 1, \dots, n \quad (21)$$

where the gain matrix $K_1 = \text{diag}\{k_{11}, \dots, k_{1n}\} > 0$, $K_2 = \text{diag}\{k_{21}, \dots, k_{2n}\} > 0$, and $\sigma_{2i} > 0$ are positive constants for improving the robustness. $\hat{\theta}_2 = [\hat{\theta}_{21}, \dots, \hat{\theta}_{2n}]^T$ and $\hat{\tau}_d$ are the estimated value of θ_2^* and τ_d , $\tilde{\theta}_2 = \hat{\theta}_2 - \theta_2^*$ and $\tilde{\tau}_d = \tau_d - \hat{\tau}_d$ are the weight error matrixes, and $A = [((k_{b1}^2 - v_{21}^2)/(k_{a1}^2 - v_{11}^2))v_{11}, \dots, ((k_{bn}^2 - v_{2n}^2)/(k_{an}^2 - v_{1n}^2))v_{1n}]^T$, $K_{v2} = [(v_{21}/(k_{b1}^2 - v_{21}^2)), \dots, (v_{2n}/(k_{bn}^2 - v_{2n}^2))]^T$. The $\hat{\tau}_d$ will be given in the following design.

The detailed design process is shown in the following steps.

Step 1: Choosing the BLF as

$$V_1 = \frac{1}{2} \sum_{i=1}^n \log \frac{k_{ai}^2}{k_{ai}^2 - v_{1i}^2} + \frac{1}{2} \sum_{i=1}^n \log \frac{k_{bi}^2}{k_{bi}^2 - v_{2i}^2} \quad (22)$$

where $k_a = [k_{a1}, \dots, k_{an}]^T$, $k_b = [k_{b1}, \dots, k_{bn}]^T$. Differentiating V_1 with respect to time yields

$$\dot{V}_1 = \sum_{i=1}^n \frac{v_{1i} \dot{v}_{1i}}{k_{ai}^2 - v_{1i}^2} + \sum_{i=1}^n \frac{v_{2i} \dot{v}_{2i}}{k_{bi}^2 - v_{2i}^2}. \quad (23)$$

To design the disturbance observer, define an assistant variable $v_3 = \tau_d + \psi(v_2)$ where $\psi(v_2)$ denotes a designed function vector. The time derivative of v_3 is

$$\begin{aligned} \dot{v}_3 &= \dot{\tau}_d + R(v_2) D_I [S(\tau) - G_k - D_k \dot{x}_{1,c} - \tau_d - \tau_e - C_k x_2] \\ &\quad - R(v_2) [\theta_2^{*T} \phi_2(Z) + \varepsilon_2 + \dot{\xi}_2 + \dot{\varphi}] \end{aligned} \quad (24)$$

where $R(v_2)$ is the derivative of $\psi(v_2)$ with respect to v_2 . For the convenience of derivation and easy implementation, $\psi(v_2)$ is chosen as a linear function with respect to v_2 . Hence, $R(v_2)$ is easily known as a constant. In order to obtain estimates of disturbance, define the estimated value of v_3 as $\hat{v}_3 = \hat{\tau}_d + \psi(v_2)$ and the derivative of $\hat{v}_3$ as follows:

$$\begin{aligned} \dot{\hat{v}}_3 &= R(v_2) D_I [S(\tau) - G_k - D_k \dot{x}_{1,c} - \hat{\tau}_d - \tau_e - C_k x_2] \\ &\quad - R(v_2) (\dot{\xi}_2 + \dot{\varphi}). \end{aligned} \quad (25)$$

The relationship between variable estimated error $\tilde{v}_3$ and disturbance estimated error $\tilde{\tau}_d$ is $\tilde{\tau}_d = \tau_d - \hat{\tau}_d = v_3 - \hat{v}_3 = \tilde{v}_3$. The time derivative of $\tilde{\tau}_d$ is as follows:

$$\dot{\tilde{\tau}}_d = \dot{\tau}_d - R(v_2) (D_I \tilde{\tau}_d + \theta_2^{*T} \phi_2(Z) + \varepsilon_2). \quad (26)$$

Thus, putting (14) and (17)–(20) into (23) results in

$$\begin{aligned} \dot{V}_1 \leq & -\sum_{i=1}^n \frac{k_{1i} v_{1i}^2}{k_{ai}^2 - v_{1i}^2} - \sum_{i=2i}^n \frac{k_{2i} v_{2i}^2}{k_{bi}^2 - v_{2i}^2} + \sum_{i=1}^n \frac{v_{2i} D_{Ii} Y(\tau)}{k_{bi}^2 - v_{2i}^2} \\ & - \sum_{i=1}^n \frac{v_{2i} \varepsilon_{2i}}{k_{bi}^2 - v_{2i}^2} - \sum_{i=1}^n \frac{v_{2i} D_{Ii} \tilde{\tau}_d}{k_{bi}^2 - v_{2i}^2} + \sum_{i=1}^n \frac{v_{2i} \tilde{\theta}_{2i}^T \phi_{2i}}{k_{bi}^2 - v_{2i}^2} \\ & + \frac{3}{4} \sum_{i=1}^n \frac{v_{2i}^2}{(k_{bi}^2 - v_{2i}^2)^2}. \end{aligned} \quad (27)$$

Step 2: For the purpose of demonstrating the system stability, define the Lyapunov function as

$$V = V_1 + \frac{1}{2} \tilde{\tau}_d^T \tilde{\tau}_d + \frac{1}{2} \sum_{i=1}^n \tilde{\theta}_{2i}^T r_{2i}^{-1} \tilde{\theta}_{2i} \quad (28)$$

where r_{2i} ($i = 1, \dots, n$) are positive constants, and the time derivative of V as follows:

$$\begin{aligned} \dot{V} \leq & -\sum_{i=1}^n \frac{k_{1i} v_{1i}^2}{k_{ai}^2 - v_{1i}^2} - \sum_{i=2i}^n \frac{k_{2i} v_{2i}^2}{k_{bi}^2 - v_{2i}^2} + \sum_{i=1}^n \frac{v_{2i} D_{Ii} Y(\tau)}{k_{bi}^2 - v_{2i}^2} \\ & - \sum_{i=1}^n \frac{v_{2i} \varepsilon_{2i}}{k_{bi}^2 - v_{2i}^2} - \sum_{i=1}^n \frac{v_{2i} D_{Ii} \tilde{\tau}_d}{k_{bi}^2 - v_{2i}^2} + \tilde{\tau}_d^T \dot{\tilde{\tau}}_d \\ & + \frac{3}{4} \sum_{i=1}^n \frac{v_{2i}^2}{(k_{bi}^2 - v_{2i}^2)^2} + \sum_{i=1}^n \frac{\tilde{\theta}_{2i}^T}{r_{2i}} \left(r_{2i} \phi_{2i} v_{2i} + \dot{\tilde{\theta}}_{2i} \right). \end{aligned} \quad (29)$$

Substituting (21) and (26) into (29)

$$\begin{aligned} \dot{V} \leq & -\sum_{i=1}^n \frac{k_{1i} v_{1i}^2}{k_{ai}^2 - v_{1i}^2} - \sum_{i=2i}^n \frac{k_{2i} v_{2i}^2}{k_{bi}^2 - v_{2i}^2} + \sum_{i=1}^n \frac{v_{2i} D_{Ii} Y(\tau)}{k_{bi}^2 - v_{2i}^2} \\ & - \sum_{i=1}^n \frac{v_{2i} \varepsilon_{2i}}{k_{bi}^2 - v_{2i}^2} - \sum_{i=1}^n \frac{v_{2i} D_{Ii} \tilde{\tau}_d}{k_{bi}^2 - v_{2i}^2} - \sum_{i=1}^n \tilde{\theta}_{2i}^T \sigma_{2i} \hat{\theta}_{2i} \\ & + \tilde{\tau}_d^T \dot{\tilde{\tau}}_d - \tilde{\tau}_d^T R(v_2) (D_I \tilde{\tau}_d + \theta_2^{*T} \phi_2(Z) + \varepsilon_2) \\ & + \frac{3}{4} \sum_{i=1}^n \frac{v_{2i}^2}{(k_{bi}^2 - v_{2i}^2)^2}. \end{aligned} \quad (30)$$

Based on *Lemma 3*, there hold

$$\frac{v_{2i} D_{Ii} Y(\tau)}{k_{bi}^2 - v_{2i}^2} \leq \frac{v_{2i}^2}{4(k_{bi}^2 - v_{2i}^2)^2} + \|D_{Ii}\|^2 \|Y(\tau)\|^2 \quad (31)$$

$$-\frac{v_{2i} \varepsilon_{2i}}{k_{bi}^2 - v_{2i}^2} \leq \frac{v_{2i}^2}{4(k_{bi}^2 - v_{2i}^2)^2} + \varepsilon_{2i}^2 \quad (32)$$

$$-\frac{v_{2i} D_{Ii} \tilde{\tau}_d}{k_{bi}^2 - v_{2i}^2} \leq \frac{v_{2i}^2}{4(k_{bi}^2 - v_{2i}^2)^2} + \|D_{Ii}\|^2 \tilde{\tau}_d^T \tilde{\tau}_d \quad (33)$$

$$-\sum_{i=1}^n \tilde{\theta}_{2i}^T \sigma_{2i} \hat{\theta}_{2i} \leq \sum_{i=1}^n \frac{\sigma_{2i}}{2} (\theta_{2i}^{*T} \theta_{2i} - \tilde{\theta}_{2i}^T \tilde{\theta}_{2i}) \quad (34)$$

$$\tilde{\tau}_d^T \dot{\tilde{\tau}}_d \leq \frac{1}{2} \tilde{\tau}_d^T \tilde{\tau}_d + \frac{1}{2} \dot{\tilde{\tau}}_d^T \dot{\tilde{\tau}}_d \quad (35)$$

$$-\tilde{\tau}_d^T R(v_2) \theta_2^{*T} \phi_2 \leq \frac{1}{2} \tilde{\tau}_d^T \|R(v_2)\|^2 \tilde{\tau}_d + \frac{1}{2} \|\phi_2\|^2 \|\theta_2^*\|^2 \quad (36)$$

$$-\tilde{\tau}_d^T R(v_2) \varepsilon_2 \leq \frac{1}{2} \tilde{\tau}_d^T \|R(v_2)\|^2 \tilde{\tau}_d + \frac{1}{2} \|\varepsilon_2\|^2. \quad (37)$$

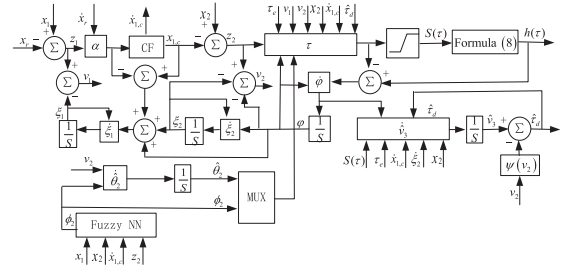


Fig. 2. Algorithm flowchart of the proposed method.

Substituting (31)–(37) into (30), and based on *Lemma 2*, the following inequality can be obtained:

$$\begin{aligned} \dot{V} \leq & -\sum_{i=1}^n k_{1i} \frac{v_{1i}^2}{k_{ai}^2 - v_{1i}^2} - \sum_{i=1}^n k_{2i} \frac{v_{2i}^2}{k_{bi}^2 - v_{2i}^2} + \frac{1}{2} \tilde{\tau}_d^T \dot{\tilde{\tau}}_d \\ & - \tilde{\tau}_d^T [R(v_2) D_I - (\|R(v_2)\|^2 + \frac{1}{2}) I - \sum_{i=1}^n \|D_{Ii}\|^2 I] \tilde{\tau}_d \\ & + \frac{3}{2} \sum_{i=1}^n \varepsilon_{2i}^2 - \sum_{i=1}^n \frac{\sigma_{2i}}{2} \tilde{\theta}_{2i}^T \tilde{\theta}_{2i} + \sum_{i=1}^n \frac{\sigma_{2i} + \|\phi_2\|^2}{2} \theta_{2i}^{*T} \theta_{2i} \\ & + \|D_{Ii}\|^2 \|Y(\tau)\|^2 \\ \leq & -\sum_{i=1}^n k_{1i} \log \frac{k_{ai}^2}{k_{ai}^2 - v_{1i}^2} - \sum_{i=1}^n k_{2i} \log \frac{k_{bi}^2}{k_{bi}^2 - v_{2i}^2} + \frac{1}{2} \dot{\tilde{\tau}}_d^T \tilde{\tau}_d \\ & - \tilde{\tau}_d^T [R(v_2) D_I - (\|R(v_2)\|^2 + \frac{1}{2}) I - \sum_{i=1}^n \|D_{Ii}\|^2 I] \tilde{\tau}_d \\ & + \frac{3}{2} \sum_{i=1}^n \varepsilon_{2i}^2 - \sum_{i=1}^n \frac{\sigma_{2i}}{2} \tilde{\theta}_{2i}^T \tilde{\theta}_{2i} + \sum_{i=1}^n \frac{\sigma_{2i} + \|\phi_2\|^2}{2} \theta_{2i}^{*T} \theta_{2i} \\ & + \|D_{Ii}\|^2 \|Y(\tau)\|^2 \leq -aV + b \end{aligned} \quad (38)$$

where

$$\begin{aligned} a = & \min \left\{ \min_{i=1, \dots, n} 2k_{1i}, \min_{i=1, \dots, n} 2k_{2i}, \lambda_{\min} \{ 2[R(v_2) D_I \right. \\ & \left. - (\|R(v_2)\|^2 + \frac{1}{2}) I - \sum_{i=1}^n \|D_{Ii}\|^2 I] \}, \min_{i=1, \dots, n} \sigma_{2i} \right\} \\ b = & \sum_{i=1}^n \|D_{Ii}\|^2 \|Y(\tau)\|^2 + \frac{3}{2} \sum_{i=1}^n \varepsilon_{2i}^2 + \sum_{i=1}^n \frac{\sigma_{2i} + \|\phi_2\|^2}{2} \theta_{2i}^{*T} \theta_{2i} \\ & + \frac{1}{2} \dot{\tilde{\tau}}_d^T \tilde{\tau}_d. \end{aligned} \quad (39)$$

Remark 2: To ensure the closed-loop stability, controller parameters should satisfy $k_{1i} > 0$, $k_{2i} > 0$, $\sigma_{2i} > 0$, and $\lambda_{\min} \{ [R(v_2) D_I - (\|R(v_2)\|^2 + (1/2)) I - \sum_{i=1}^n \|D_{Ii}\|^2 I] \} > 0$. Therefore, $\dot{V}$ will be negative definite. It can be known that V is bounded for $\forall t > 0$.

The algorithm flowchart of the proposed scheme is shown in Fig. 2.

Then, the following theorem is proposed.

Theorem 1. For the robotic manipulator system (1), if the command filter is designed according to *Lemma 4*, with the disturbance observer (25), the auxiliary system (15), the virtual control law (17), the control input vector (18) with compensation signals (19) and (20), and adaptive laws (21), given that initial conditions satisfy $v_{1i}(0) \in \Omega_{v1}$ and $v_{2i}(0) \in \Omega_{v2}$. It can

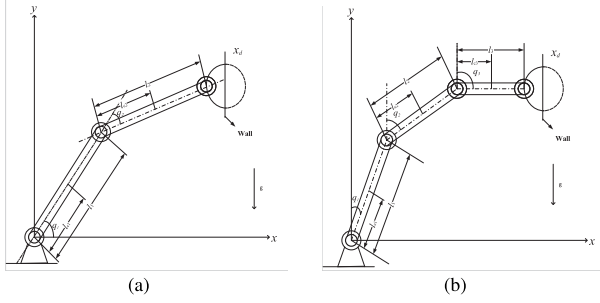


Fig. 3. (a) 2-DOF planar robotic manipulator. (b) 3-DOF redundant planar robotic manipulator.

TABLE I
PARAMETERS OF THE 2-DOF ROBOTIC MANIPULATOR

Parameter	link 1	link 2
$l_i(\text{m})$	1.40	1.24
$l_{ci}(\text{m})$	0.70	0.62
$m_i(\text{kg})$	2	0.85
$I_i(\text{kgm}^2)$	0.98	0.953

be concluded that v_{1i} , v_{2i} , ζ_{1i} , ζ_{2i} , and θ_{2i} in the close-loop system are semiglobally uniformly ultimately bounded (SGUUB) and will converge into the arbitrarily small regions, all the states will be bounded and the full-state constraints are not violated. Meanwhile, x_{1i} and x_{2i} will remain in the set Ω_{x1} and Ω_{x2} , respectively.

Proof. See Appendix.

IV. SIMULATION RESULT

Three set comparisons are listed to illustrate the effectiveness of the proposed control method by simulating the control virtual robotic manipulators, as shown in Fig. 3. Fig. 3(a) shows a typical 2-DOF nonredundant robotic manipulator. Fig. 3(b) shows a 3-DOF redundant planar robotic manipulator. Let m_i and l_i denote the mass and length of the link i , respectively. Let l_{ci} denote the distance from joint $i - 1$ of the robotic manipulator to the centroid of the link i . Let I_i denote the moment of inertia of link i about an axis coming out of the page passing through the center of mass of link i , ($i = 1, \dots, n$).

A. Controlling a Nonredundant Robotic Manipulator

Parameters of the 2-DOF robotic manipulator system are defined in Table I [37].

The dynamics can be expressed as (2). The matrices D_k , C_k , and G_k are defined as [22]

$$D_k = J^{-T} D_q J^{-1}, \quad C_k = J^{-T} (C_q - D_q J^{-1} \dot{J}) J^{-1}$$

$$G_k = J^{-T} G_q.$$

D_q , C_q , G_q , and J are defined as

$$D_q = \begin{bmatrix} D_{q11} & D_{q12} \\ D_{q21} & D_{q22} \end{bmatrix}, \quad C_q = \begin{bmatrix} C_{q11} & C_{q12} \\ C_{q21} & 0 \end{bmatrix}, \quad G_q = \begin{bmatrix} G_{q11} \\ G_{q21} \end{bmatrix}$$

$$J = \begin{bmatrix} -(l_1 \sin q_1 + l_2 \sin(q_1 + q_2)) & -l_2 \sin(q_1 + q_2) \\ l_1 \cos q_1 + l_2 \cos(q_1 + q_2) & l_2 \cos(q_1 + q_2) \end{bmatrix}$$

where $D_{q11} = m_1 l_{c1}^2 + m_2 (l_1^2 + l_{c2}^2 + 2l_1 l_{c2} \cos q_2) + I_1 + I_2$, $D_{q21} = D_{q12} = m_2 (l_{c2}^2 + l_1 l_{c2} \cos q_2) + I_2$,

$$D_{q22} = m_2 l_{c2}^2 + I_2, \quad C_{q11} = -m_2 l_1 l_{c2} \dot{q}_2 \sin q_2,$$

$$C_{q12} = -m_2 l_1 l_{c2} (\dot{q}_1 + \dot{q}_2) \sin q_2, \quad C_{q21} = m_2 l_1 l_{c2} \dot{q}_1 \sin q_2,$$

$$G_{q11} = (m_1 l_{c1} + m_2 l_1) g \cos q_1 + m_2 l_{c2} g \cos(q_1 + q_2), \quad \text{and}$$

$$G_{q21} = m_2 l_{c2} g \cos(q_1 + q_2).$$

The initial parameters of the robotic manipulator are $x_1(0) = [0.5, 0.7]^T \text{m}$ and $\dot{x}_1(0) = x_2(0) = [0, 0]^T \text{m/s}$. The initial value of the compensation signals are defined as $\zeta_1 = \zeta_2 = [0, 0]^T$. Chose $t \in [0, 10]$. The commanded trajectory $x_d(t)$ is selected as $x_d(t) = [x_{d1}, x_{d2}]^T = [0.7 - 0.2 \cos(t), 0.7 + 0.2 \sin(t)]^T$. Define the unknown disturbance $\tau_d = [10 \sin(t), 6 \sin(t)]^T$. The wall is 0.8 m away from the coordinate origin along the y-axis ($x = 0.8$), and when the end-effector contacts the wall, it will move along the solid wall. Select $M_d = \text{diag}[2, 2]$, $B_d = \text{diag}[20, 20]$, and $K_d = \text{diag}[50, 50]$. The parameters of the command filter are $\omega_n = 1000$, $\zeta = 3$. Chose $R(v_2) = \text{diag}[30, 30]$. The number of fuzzy rules is 8, $c_{ik} \in [-4, 3]$, $c_{ik} \in N$ and $\sigma_{ik}^2 = 2$ for $k = 1, \dots, 8$. The initial values of $\hat{\theta}_{21}$ and $\hat{\theta}_{22}$ are set to 0.

To verify the effectiveness of the proposed control method, the proposed control method is compared with adaptive fuzzy NN command filtered impedance control with full-state constraint, which does not consider unknown disturbance and input saturation. Detailed control parameters are given in Case 1.

Case 1. For the proposed control method in this article, select $K_1 = \text{diag}[20, 28]$, $K_2 = \text{diag}[120, 120]$, $K_3 = [320, 320]^T$, $k_a = [0.0001, 0.0001]^T$, $k_b = [0.03, 0.03]^T$, $r_{2i} = 20$, and $\sigma_{2i} = 0.01$ ($i = 1, 2$). And the amplitudes of the input saturation are chosen as $\tau_{i \max} = 60$ and $\tau_{i \min} = -60$; For adaptive NN-based command filtered impedance control with full-state constraint, select $K_1 = \text{diag}[20, 28]$, $K_2 = \text{diag}[120, 120]$, $k_a = [0.0001, 0.0001]^T$, $k_b = [0.03, 0.03]^T$, $r_{2i} = 20$, and $\sigma_{2i} = 0.01$ ($i = 1, 2$).

The simulation results are described in Figs. 4 and 5. Fig. 4(a)–(d) shows that the proposed control method can achieve effective tracking and is more stable than the control method without considering the unknown disturbance. It can be known from Fig. 4(c) and (d) that the proposed method makes v_1 and v_2 converge to a small region around zero if initial states satisfy $v_{1i}(0) \in \Omega_{v1}$ and $v_{2i}(0) \in \Omega_{v2}$ and cannot violate the state constraints. Fig. 4(e) and (f), respectively, shows the control input τ with and without input saturation. It can be known from Fig. 4(e) that the control input of the proposed control method satisfies the input saturation condition. Fig. 5(a) and (b) shows that the disturbance observer can track time-varying disturbance effectively, and the estimated errors are converging to a small neighborhood of zero. Fig. 5(c) and (d), respectively, presents the two-norm of adaptive law $\hat{\theta}_{2i}$ in the two control methods, which are bounded by an unknown constant.

For giving a better view of the proposed control method, another simulation experiment compares it with adaptive fuzzy NN command filtered impedance control with disturbance observer, which does not consider state constraints and input saturation. Detailed control parameters are given in Case 2.

Case 2. For the proposed control method in this article, the control parameters are selected as the same as those in case 1, and the simulation results are the same as in case 1.

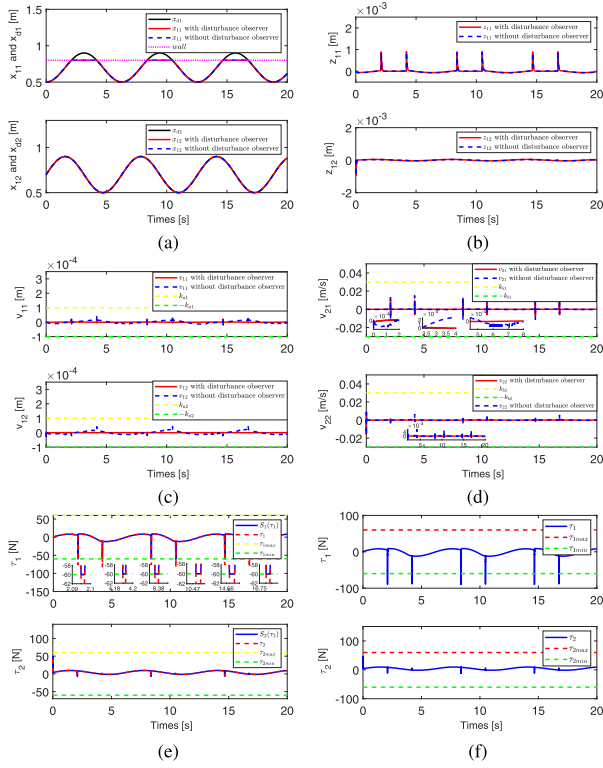


Fig. 4. Simulation results in case 1. (a) Output signal x_1 . (b) Tracking error z_1 . (c) Tracking error v_1 . (d) Tracking error v_2 . (e) Control force τ with input saturation. (f) Control force τ without input saturation.

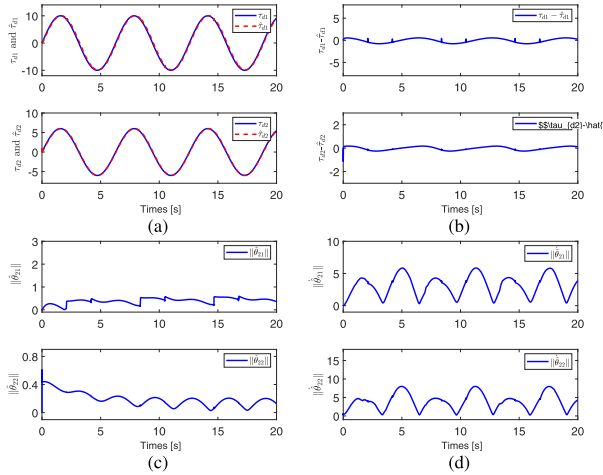


Fig. 5. Performance of disturbance observer and adaptive law $\hat{\theta}_2$ in case 1. (a) Disturbance estimation tracking. (b) Disturbance estimate errors. (c) Adaptive law $\hat{\theta}_2$ of the control method proposed in this article. (d) Adaptive law $\hat{\theta}_2$ of the control method without disturbance observer and input saturation.

For adaptive NN-based command filtered impedance control with disturbance observer, select $K_1 = \text{diag}[30, 38]$, $K_2 = \text{diag}[120, 120]$, $r_{2i} = 20$, and $\sigma_{2i} = 0.01 (i = 1, 2)$.

The simulation results are described in Figs. 6 and 7. Fig. 6(a)–(d) shows that the proposed control method can achieve effective tracking, is more stable than the comparative control method, and does not violate the constraints. Fig. 6(e) shows the control input τ of the control method without considering input saturation. Fig. 6(f) presents the two-norm of adaptive law $\hat{\theta}_{2i}$ in the control method without state constraint and input saturation which are bounded by an unknown constant. Fig. 7(a) and (b) shows that disturbance observer

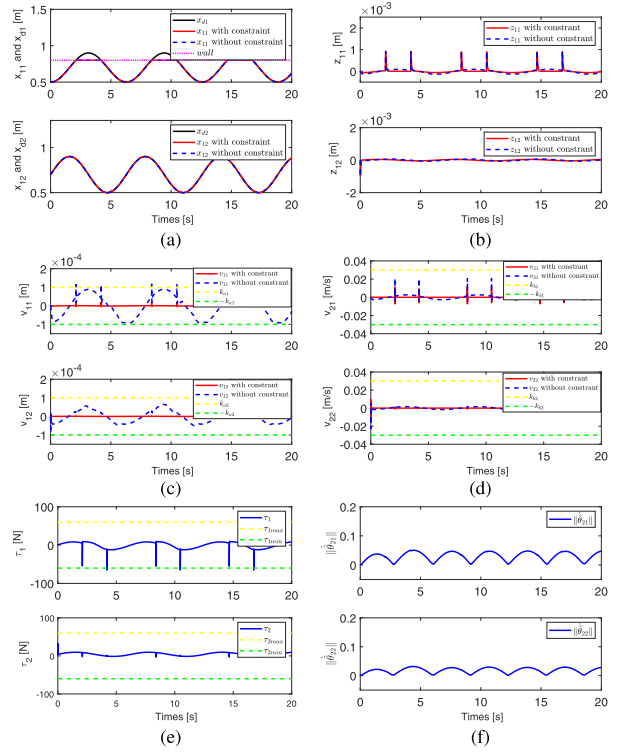


Fig. 6. Simulation results of case 2. (a) Output signal x_1 . (b) Tracking error z_1 . (c) Tracking error v_1 . (d) Tracking error v_2 . (e) Control force τ with input saturation. (f) Adaptive law $\hat{\theta}_2$ of the control method without state constraint.

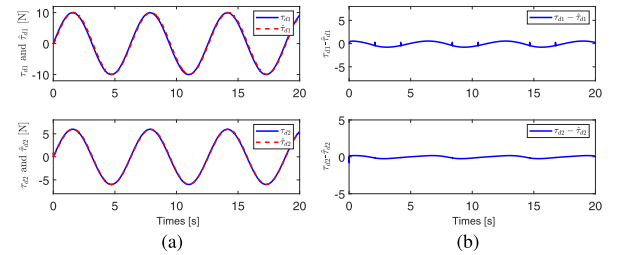


Fig. 7. Performance of disturbance observer in case 2. (a) Disturbance estimation tracking. (b) Disturbance estimate errors.

can track time-varying disturbance effectively, the estimated errors are converging to a small neighborhood of zero.

B. Controlling a Redundant Robotic Manipulator

Parameters of the 3-DOF robotic manipulator system are given in Table II [38]. Then, the inverse relations of (2) for a redundant manipulator can be expressed as

$$\dot{q} = J^+ \dot{x} + \dot{\theta}_N, \ddot{q} = J^+ (\ddot{x} - \dot{J} \dot{q}) + \ddot{\theta}_N$$

where $\dot{\theta}_N$ and $\ddot{\theta}_N$ are the vectors of joints velocity and acceleration in the null space of J . In [38], an error signals $\dot{e}_N = (I - J^+ J) \dot{h} - \dot{\theta}_N$ and $\ddot{e}_N = -K_N \dot{e}_N$, where $K_N = \text{diag}[k_{N1}, \dots, k_{Nm}]$ is a positive definite design matrix and $h(\cdot) \in \mathbb{R}^m$ is a given vector function and is the gradient of the objective function $H(q)$, were designed to make sure that the main task control performance is not affected, while the subtask is also achieved.

In this simulation, the given subtask is selected as minimizing joint motions, which means minimizing $\|\dot{q}\|$. For realizing this subtask, $H(q)$ is chosen to be zero, the null space velocity, and $(I - J^+ J) \dot{h}$ will go to zero.

TABLE II
PARAMETERS OF THE 3-DOF ROBOTIC MANIPULATOR

Parameter	link 1	link 2	link 3
$l_i(\text{m})$	0.268	0.258	0.268
$l_{ci}(\text{m})$	0.134	0.128	0.095
$m_i(\text{kg})$	0.41	4.10	0.095
$I_i(\text{kgm}^2)$	4.52×10^{-3}	6.011×10^{-2}	4.52×10^{-3}

The dynamics can be expressed as (4). The matrices D_k , C_k , and G_k are defined as [38]

$$D_k = J^{-T} D_q J^{-1}, \quad C_k = J^{-T} (C_q - D_q J^{-1} \dot{J}) J^{-1}$$

$$G_k = J^{-T} G_q.$$

D_q , C_q , G_q , and J are defined as

$$D_q = \begin{bmatrix} p_{11} & p_{12} \cos(q_2 - q_1) & p_{13} \cos(q_3 - q_1) \\ p_{12} \cos(q_2 - q_1) & p_{22} & p_{23} \cos(q_3 - q_2) \\ p_{13} \cos(q_3 - q_1) & p_{23} \cos(q_3 - q_2) & q_{33} \end{bmatrix}$$

$$C_q = \begin{bmatrix} 0 & -p_{12} \dot{q}_2 \sin(q_2 - q_1) & -p_{13} \dot{q}_3 \sin(q_3 - q_1) \\ p_{12} \dot{q}_1 \sin(q_2 - q_1) & 0 & -p_{23} \dot{q}_3 \sin(q_3 - q_2) \\ p_{13} \dot{q}_1 \sin(q_3 - q_1) & p_{23} \dot{q}_2 \sin(q_3 - q_2) & 0 \end{bmatrix}$$

$$G_q = \begin{bmatrix} -(m_1 l_{c1} + m_2 l_1 + m_3 l_1) g \sin(q_1) \\ -(m_2 l_{c2} + m_3 l_2) g \sin(q_2) \\ -m_3 l_{c3} g \sin(q_3) \end{bmatrix}$$

$$J = \begin{bmatrix} j_{11} + j_{12} + j_{13} & j_{12} + j_{13} & j_{13} \\ j_{21} + j_{22} + j_{23} & j_{22} + j_{23} & j_{23} \end{bmatrix}$$

where $p_{11} = (m_2 + m_3)l_1^2 + m_1 l_{c1}^2 + I_1$, $p_{22} = m_2 l_{c2}^2 + m_3 l_2^2 + I_2$, $p_{33} = m_3 l_{c3}^2 + I_3$, $p_{12} = (m_2 l_{c2} + m_3 l_2)l_1$, $p_{13} = m_3 l_1 l_{c3}$, $p_{23} = m_3 l_2 l_{c3}$, $j_{11} = -l_1 \sin(q_1)$, $j_{12} = -l_2 \sin(q_1 + q_2)$, $j_{13} = -l_3 \sin(q_1 + q_2 + q_3)$, $j_{21} = l_1 \cos(q_1)$, $j_{22} = l_2 \cos(q_1 + q_2)$, and $j_{23} = l_3 \cos(q_1 + q_2 + q_3)$.

The initial parameters of the robotic manipulator are $x_1(0) = [0, 0.258]^T \text{m}$ and $\dot{x}_1(0) = x_2(0) = [0, 0]^T \text{m/s}$. The initial value of the compensation signals are defined as $\xi_1 = \xi_2 = [0, 0]^T$. $\dot{\theta}_N(0) = [1, 3.5, 1.5]^T$. Chose $t \in [0, 10]$. The commanded trajectory $x_d(t)$ is selected as $x_d(t) = [x_{d1}, x_{d2}]^T = [0.258 + 0.258 \cos(t + \pi), 0.258 + 0.258 \sin(t)]^T$. Define the unknown disturbance moment vector as $\tau_d = [4 \sin(t), 5 \sin(t)]^T$. The wall is 0.387 m away from the coordinate origin along the y -axis ($x = 0.387$), and when the manipulator end-effector contacts the wall, it will move along the solid wall. Selected $M_d = \text{diag}[2, 2]$, $B_d = \text{diag}[10, 10]$, and $K_d = \text{diag}[50, 50]$. The parameters of the command filter are $\omega_n = 1000$ and $\zeta = 3$. Chose $R(v_2) = \text{diag}[30, 30]$ and $K_N = \text{diag}[2, 3, 4]$. The number of fuzzy rules is 7, $c_{ik} \in [-3, -3]$, $c_{ik} \in N$, and $\sigma_{ik}^2 = 2$, for $k = 1, \dots, 7$. The initial position of θ_1 and θ_2 are all zero.

In order to verify the effectiveness of the proposed control method for redundant manipulators, the proposed control method is compared with adaptive NN-based command filtered impedance control with full-state constraint, which does not consider unknown disturbance and input saturation. The detailed control parameters will be given in Case 3.

Case 3. For the control method proposed in this article, the control parameters are selected as $K_1 = \text{diag}[120, 128]$,

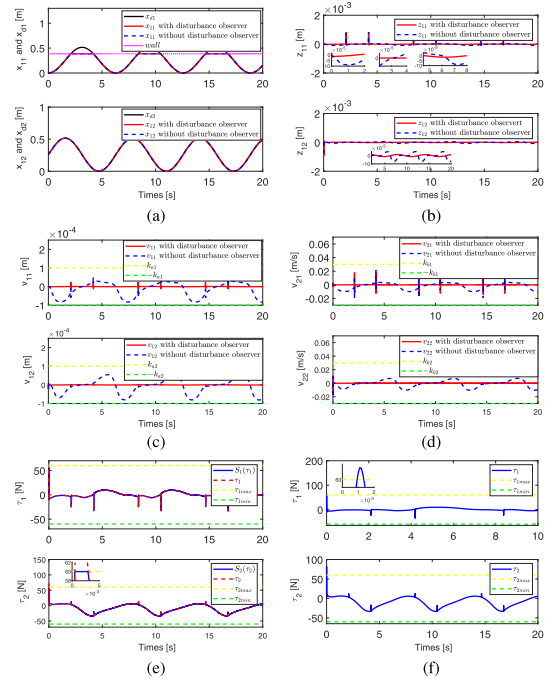


Fig. 8. Simulation results of case 3. (a) Output signal x_1 . (b) Tracking error z_1 . (c) Tracking error v_1 . (d) Tracking error v_2 . (e) Control force τ with input saturation. (f) Control force τ without input saturation.

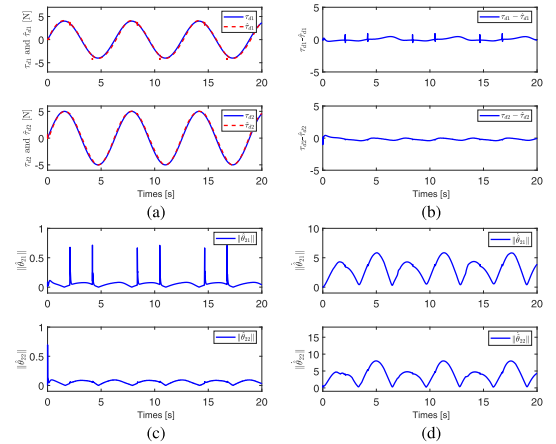


Fig. 9. Performance of disturbance observer and adaptive law $\hat{\theta}_2$ in case 3. (a) Disturbance estimation tracking. (b) Disturbance estimate errors. (c) Adaptive law $\hat{\theta}_2$ of the control method proposed in this article. (d) Adaptive law $\hat{\theta}_2$ of the control method without disturbance observer and input saturation.

$K_2 = \text{diag}[160, 400]$, $K_3 = [120, 160]^T$, $k_a = [0.0001, 0.0001]^T$, $k_b = [0.03, 0.03]^T$, $r_{2i} = 20$, and $\sigma_{2i} = 1 (i = 1, 2)$. And the amplitudes of the input saturation are chosen as $\tau_{i\max} = 60$ and $\tau_{i\min} = -60$. For adaptive NN-based command filtered impedance control with full-state constraint, the control parameters are selected as $K_1 = \text{diag}[120, 128]$, $K_2 = \text{diag}[160, 400]$, $k_a = [0.0001, 0.0001]^T$, $k_b = [0.03, 0.03]^T$, $r_{2i} = 20$, and $\sigma_{2i} = 1 (i = 1, 2)$.

The simulation results are described in Figs. 8–10. Among them, Fig. 8(a)–(d) shows that the proposed control method can achieve effective tracking and is more stable than the control method without considering the unknown disturbance. It can be known from Fig. 8(c) and (d) that the proposed method makes v_1 and v_2 converge to a small region around zero if initial states satisfy $v_{1i}(0) \in \Omega_{v1}$ and $v_{2i}(0) \in \Omega_{v2}$.

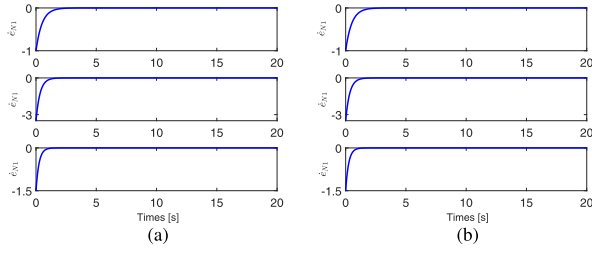


Fig. 10. Subtask tracking errors in case 3. (a) Subtask tracking errors with the proposed method. (b) Subtask tracking errors with the control method without disturbance observer and input saturation.

and cannot violate the state constraints. Fig. 8(e) and (f), respectively, shows the control input τ with and without input saturation. It can be known from Fig. 8(e) that the control input of the proposed control method satisfies the input saturation condition. Fig. 9(a) and (b) shows that disturbance observer can track time-varying disturbance effectively, and the estimated errors are converging to a small neighborhood of zero. Fig. 9(c) and (d), respectively, presents the two-norm of adaptive law $\hat{\theta}_{2i}$ in the two control methods which are bounded by an unknown constant.

Remark 3: Subtask tracking errors are shown in Fig. 10, from which it can be known that the given subtask is finished, while the main task execution has not been affected.

V. CONCLUSION

In this article, an adaptive fuzzy NN command filtered impedance control with disturbance observer has been proposed for constrained robotic manipulators in an unknown environment. Realize the full-state constraints by combining BLFs with the backstepping method, and an adaptive fuzzy NN control is used to handle the unknown system dynamics. By designing a disturbance observer to eliminate the effect of the unknown disturbance, an auxiliary system is designed to hand the input saturation effect. The CFC technology and error compensation mechanism were introduced to improve the control accuracy. The simulation results prove the effectiveness of the proposed method.

APPENDIX

Further analysis can get: $V(t) \leq (V(0) - (\rho/\mu))e^{-\mu t} + (\rho/\mu)$. According to *Lemma 1*, it can be known that all the signals in the closed-loop system can be semiglobally uniformly ultimately bounded (SGUUB) and can further know that

$$\begin{cases} \frac{1}{2} \log \frac{k_{ai}^2}{k_{ai}^2 - v_{1i}^2} \leq V \leq \left(V(0) - \frac{\rho}{\mu}\right)e^{-\mu t} + \frac{\rho}{\mu} \\ \frac{1}{2} \log \frac{k_{bi}^2}{k_{bi}^2 - v_{2i}^2} \leq V \leq \left(V(0) - \frac{\rho}{\mu}\right)e^{-\mu t} + \frac{\rho}{\mu} \end{cases} \quad (40)$$

can be solved as: $|v_{1i}| \leq k_{ai}(1 - e^{-2[(V(0) - (\rho/\mu))e^{-\mu t} + (\rho/\mu)]})^{1/2} < k_{ai}$ and $|v_{2i}| \leq k_{bi}(1 - e^{-2[(V(0) - (\rho/\mu))e^{-\mu t} + (\rho/\mu)]})^{1/2} < k_{bi}$.

In order to prove that ξ_1 , ξ_2 and φ can converge to the suitable region in a finite time, the following Lyapunov function is selected:

$$\bar{V} = \frac{1}{2}\xi_1^T \xi_1 + \frac{1}{2}\xi_2^T \xi_2 + \frac{1}{2}\varphi^T \varphi. \quad (41)$$

According to *Lemma 4*, it can be known that $|x_{1,c} - \alpha| \leq \beta = [\beta_1, \dots, \beta_n]^T$. The time derivative of $\bar{V}$ can be written as follows:

$$\begin{aligned} \dot{\bar{V}} \leq & -\xi_1^T K_1 \xi_1 + \xi_1^T \xi_2 + \xi_1^T \varphi + \xi_1^T \beta - \xi_2^T K_2 \xi_2 - \xi_2^T K_2 \varphi \\ & - \varphi^T D_I K_\eta \varphi + \varphi^T D_I \Delta \tau + \varphi^T D_I Y(\tau). \end{aligned} \quad (42)$$

Based on *Lemma 3*, (42) can be written as follows:

$$\begin{aligned} \dot{\bar{V}} \leq & -\xi_1^T \left(K_1 - \frac{3}{2}I\right) \xi_1 - \xi_2^T \left(\frac{1}{2}K_2 - \frac{1}{2}I\right) \xi_2 \\ & - \varphi^T \left(D_I K_\eta - \frac{3}{2}I - \frac{1}{2}K_2\right) \varphi + \frac{1}{2}\beta^T \beta \\ & + \frac{1}{2}\|D_I\|^2 \|\Delta \tau\|^2 + \frac{1}{2}\|D_I\|^2 \|Y(\tau)\|^2 \\ \leq & -a_0 \bar{V} + b_0 \end{aligned} \quad (43)$$

where

$$\begin{aligned} a_0 = & \min\{\lambda_{\min}(2K_1 - 3I), \lambda_{\min}(K_2 - I), \lambda_{\min}(2D_I K_\eta \\ & - 3I - K_2)\} \\ b_0 = & \frac{1}{2}\beta^T \beta + \frac{1}{2}\|D_I\|^2 \|\Delta \tau\|^2 + \frac{1}{2}\|D_I\|^2 \|Y(\tau)\|^2. \end{aligned} \quad (44)$$

To ensure the closed-loop stability, controller parameters should satisfy $\lambda_{\min}(2K_1 - 3I) > 0$, $\lambda_{\min}(K_2 - I) > 0$, and $\lambda_{\min}(2D_I K_\eta - 3I - K_2) > 0$. According to *Lemma 1*, from the above-mentioned equation, it is easy to obtain that ξ_1 , ξ_2 , and φ are SGUUB and

$$\lim_{t \rightarrow \infty} |\xi_{ji}| \leq \sqrt{\frac{2b_0}{a_0}}, \lim_{t \rightarrow \infty} |\varphi_i| \leq \sqrt{\frac{2b_0}{a_0}}, j = 1, 2; i = 1, \dots, n. \quad (45)$$

Since $v_1 = z_1 - \xi_1$, then $|z_{1i}| \leq |v_{1i}| + |\xi_{1i}| \leq k_{ai} + ((2b_0/a_0))^{1/2}$. And $x_1 = z_1 + x_r(t)$, $|x_r(t)| \leq Y_0$, it can be concluded that $|x_{1i}| \leq |z_{1i}| + |x_{ri}| \leq k_{ai} + ((2b_0/a_0))^{1/2} + Y_0 \leq k_{c1i}$. According to (10), it can be realized that α is a function of $z_1, \dot{x}_r$. Thus, the supremum s_1 of α holds. Due to $|x_{1,c} - \alpha| < \beta$, we have $|x_{1,c}| \leq \beta_i + s_{1i} \leq \chi_i$. By $|z_{2i}| \leq k_{bi} + 2((2b_0/a_0))^{1/2}$ and $z_2 = x_2 - x_{1,c}$, then $|x_{2i}| \leq k_{bi} + 2((2b_0/a_0))^{1/2} + \chi_i \leq k_{c2i}$. Now, it can be determined that v_i, z_i, x_i , and τ are bounded. The proof is completed.

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